No Finite Separable Representation at Three Prime Generators

Nonsingular minors of every order, two-prime Hecke–Mahler polynomials, and the open boundary for Erdős Problem #269

Will Cook*

July 2026

Abstract

For three pairwise distinct primes, the reciprocal running-LCM kernel has nonsingular minors of every order, with one choice of indices valid in every third-coordinate layer. Dividing out row and column powers leaves a binary carry whose consecutive threshold columns differ on disjoint intervals. This gives both arbitrary-order non-separability and the exact rank of every finite sample. Steve Fan posted the two-prime factorisation, reduction and transcendence conclusion first (erdosproblems.com/269, 26 June 2026, post-7218). This paper records that argument as an ordinary deduction from Bugeaud–Laurent and does not claim priority or a Lean theorem for the two-prime case. The repeated three-prime irrationality question remains open.

1 Introduction

Let P be a finite set of at least two primes and $\mathcal{S}_P$ its positive smooth integers. Put

$$\mathrm{L}(x) = \mathrm{lcm}\{u \in \mathcal{S}_P : u \le x\}, \qquad \mathcal{R}_P = \sum_{u \in \mathcal{S}_P} \mathrm{L}(u)^{-1}.$$

Erdős asks whether $\mathcal{R}_P$ is irrational [1, p. 65][2, p. 106]. For $P = \{p, q, r\}$, define

$$\mathrm{H}(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor}, \qquad \mathrm{K}(i, j, k) = \mathrm{H}(p^i q^j r^k)^{-1}.$$

Prime-power divisibility gives $\mathrm{L} = \mathrm{H}$ (Proposition 4.1).

With two primes, the corresponding kernel factors into a function of each exponent separately, reducing the double sum to a product. The first question at three primes is whether finitely many such products still suffice. Theorem 2.1 rules this out: one floor carry survives after all row and column factors have been removed. We first explain that obstruction, then recover the two-prime comparison, and finally identify the arithmetic of the actual three-prime tails. The last step is necessary because the rank of a kernel does not determine the irrationality of its sum.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

2 The binary carry and arbitrary-order rank

Theorem 2.1 (arbitrary-order non-separability)**.** *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 0$ there are injective maps $I, J : \{0, \dots, n-1\} \to \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(\mathrm{K}(I(a), J(b), k))_{0 \leq a,b<n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell(i)$ and $G_\ell(j,k)$, $0 \leq \ell < d$, satisfying

$$\mathrm{K}(i,j,k) = \sum_{\ell<d} f_\ell(i) G_\ell(j,k) \qquad \text{for all } i,j,k.$$

Proof. Put $c = r^{-1}$, $x_i = \{i \log_r p\}$ and $y_j = \{j \log_r q\}$. Splitting the floor exponents gives

$$\mathrm{K}(i,j,k) = U_i(k)^{-1} C_{ij} V_j(k)^{-1}, \qquad C_{ij} = c^{\mathbf{1}_{\{x_i + y_j \geq 1\}}},$$

where

$$U_i(k) = p^i q^{\lfloor \log_q(p^i r^k) \rfloor} r^{k + \lfloor \log_r p^i \rfloor},$$
$$V_j(k) = q^j p^{\lfloor \log_p(q^j r^k) \rfloor} r^{\lfloor \log_r q^j \rfloor}.$$

Both factors are positive. Distinct primes make $\log_r p$ and $\log_r q$ irrational, so each individual rotation is dense. For $n \geq 1$, choose distinct row indices with $0 < x_{I(0)} < \cdots < x_{I(n-1)} < 1$. Choose $1 - y_{J(0)}$ in $(0, x_{I(0)})$, and, for $1 \leq b < n$, choose $1 - y_{J(b)}$ in $(x_{I(b-1)}, x_{I(b)})$. The intervals are disjoint, so the column indices are distinct. These choices give

$$T_n(c) = \begin{pmatrix} c & 1 & \cdots & 1 \\ c & c & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ c & c & \cdots & c \end{pmatrix}, \qquad \det T_n(c) = c(c-1)^{n-1}.$$

Indeed, subtracting each preceding row from the next, working upwards from the last row, leaves diagonal entries $c, c-1, \dots, c-1$. The same I, J work for every k, whose only effect is multiplication by nonzero row and column factors. The empty minor for $n = 0$ equals 1. Finally, d separated summands would factor every $(d+1) \times (d+1)$ restriction through a d-dimensional space, contradicting the nonzero minor. □

Both clauses are uniform rank and nonseparation.

Corollary 2.2 (the same minors modulo every admissible denominator)**.** *For $(p,q,r) = (2,3,5)$, the maps I, J in Theorem 2.1 can be chosen so that their minors are invertible over $\mathbb{Z}/B\mathbb{Z}$ for every $B \geq 2$ coprime to 30 and every $k \geq 0$.*

Proof. Every row and column factor is a unit modulo B. The normalised determinant is $5^{-1}(-4/5)^{n-1}$, also a unit. The choices of I, J are independent of both B and k. □

The statement is admissible modular minors.

Example 2.3. For $(p,q,r) = (2,3,5)$, the leading two-by-two determinant is $-1/15$: its four entries are $1, 1/6, 1/2, 1/60$, so it equals $1/60 - 1/12$.

The four entries and the determinant are two by two fixture.

The indices must be selected. The leading 4×4 block at $\{2, 3, 5\}$ is singular:

$$\mathrm{K}(3, j, 0) = \frac{1}{120}\,\mathrm{K}(0, j, 0) \qquad (0 \le j < 4).$$

These equalities follow by evaluating the height at 3^j and $8 \cdot 3^j$. At $j = 4$ the difference is $-1/19440000$. Thus the leading minors do not supply the arbitrary-order theorem.

Proposition 2.4 (finite sampled cut rank)**.** *Let $m \ge 1$ and let c lie in a field with $c \ne 0, 1$. For $0 \le k \le m$ write v_k for the length-m column with a 1 in each of the first k coordinates and c thereafter. A matrix whose distinct columns are v_k for k in a nonempty set E has rank $|E| - \mathbf{1}_{\{0,m\} \subseteq E}$.*

Proof. List $E = \{k_1 < \cdots < k_t\}$. The $t-1$ differences $v_{k_{j+1}} - v_{k_j} = (1-c)\mathbf{1}_{\{k_j,\dots,k_{j+1}-1\}}$ have disjoint nonempty supports, hence are linearly independent. If $k_1 > 0$ or $k_t < m$, their union misses a coordinate where v_{k_1} is nonzero, so the rank is t. If $k_1 = 0$ and $k_t = m$, then $v_0 = c\,\mathbf{1}$ lies in the span of the differences (their sum is $(1-c)\mathbf{1}$), so the rank is $t - 1$. □

The statement is rank cut matrix.

The formula classifies every finite sampled kernel rank after the row and column factors of Theorem 2.1, independently of the third coordinate. An exact logarithm-free algorithm orders the rationals $p^i/r^{\lfloor \log_r p^i \rfloor}$, counts those strictly below each $r^{\lfloor \log_r q^j \rfloor + 1}/q^j$, and applies the displayed correction. Direct elimination on the $\{2, 3, 5\}$ kernel agrees with the formula through order 24; larger leading ranks recorded with the accompanying checker use the formula and exact integer phase comparisons.

The norm matters. For the infinite normalised carry matrix,

$$\inf_{F \text{ of finite separated rank}} \|C - F\|_\infty = (1-c)/2.$$

Distinct columns are exactly $1 - c$ apart in the supremum norm, by density of the row phases. An approximation below half that distance would give infinitely many bounded, uniformly separated columns in a finite-dimensional space. Compactness excludes this. The constant $(1 + c)/2$ attains the bound.

Finite restrictions have a different behaviour. At $c = 1/5$,

$$T = \begin{pmatrix} 1/5 & 1 \\ 1/5 & 1/5 \end{pmatrix}, \qquad A = \begin{pmatrix} 3/10 & 9/10 \\ 1/10 & 3/10 \end{pmatrix}$$

satisfy $\det A = 0$ and $\|T - A\|_{\max} = 1/10 < 2/5$. For the original kernel, truncating the first coordinate at N gives separated rank at most N and

$$\|K - K^{(N)}\|_{\ell^1} \le \frac{pqr\,p^{-3N}}{(1-p^{-3})(1-q^{-3})(1-r^{-3})}.$$

This follows from $H(x) > x^3/(pqr)$. Thus the infinite uniform obstruction coexists with geometric approximation in the summation norm. The scalar irrationality question requires arithmetic information about the actual multiplicities.

3 The two-prime comparison

For distinct primes $p < q$, write $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$. Prime-power divisibility identifies this product with the running LCM. Let $\mathcal{R}_{p,q}$ retain every smooth-number multiplicity, and let $\mathcal{D}_{p,q}$ retain the initial value and one reciprocal at each positive pure-power jump. Fan's argument gives the repeated two-prime factorisation and its quadratic reduction to a single Hecke–Mahler value. The same boundary-channel calculation also yields the de-duplicated affine formula displayed below [7].

Theorem 3.1 (both two-prime sums). *Let $p < q$ be distinct primes. Put $\theta = \log p / \log q$ and $A = \sum_{n \geq 0} p^{-n} q^{-\lfloor n\theta \rfloor}$. Let $\mathcal{R}_{p,q}$ sum the reciprocal running LCM at every positive $\{p, q\}$-smooth integer, and let $\mathcal{D}_{p,q}$ count each distinct running LCM once. Then*

$$\mathcal{D}_{p,q} = \frac{(q-p)A + p}{q-1}, \qquad \mathcal{R}_{p,q} = \frac{(p+q-1)A - (p-1)A^2}{q-1}. \tag{1}$$

Both numbers are transcendental.

Proof. Set $x = 1/p$, $y = 1/q$, $m_n = \lfloor n\theta \rfloor$ and $\delta_n = m_{n+1} - m_n$. Unique factorisation makes θ irrational, and $0 < \theta < 1$ gives $\delta_n \in \{0, 1\}$. The initial value and the p-channel contribute $A = \sum_{n \geq 0} x^n y^{m_n}$. There is one q-power between p^n and p^{n+1} precisely when $\delta_n = 1$, so the other channel contributes

$$B = \sum_{n \geq 0} \delta_n x^n y^{m_n + 1}, \qquad \mathcal{D}_{p,q} = A + B.$$

All these series converge absolutely. Since $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n}(y-1)$, shifting the series for A gives $A - 1 - xA = x(y-1)B/y$. Therefore

$$B = \frac{p - (p-1)A}{q-1}.$$

At $p^i q^j$ the height is $p^{i + \lfloor j/\theta \rfloor} q^{j + m_i}$, hence

$$\mathcal{R}_{p,q} = A \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor} = A(1 + B).$$

For the last equality, each $j \geq 1$ corresponds to $n = \lfloor j/\theta \rfloor$ with $m_n = j - 1$ and $\delta_n = 1$. These identities prove (1).

For the Hecke–Mahler series $F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{m_n} x^n y^k$, geometric summation gives

$$A = \frac{1}{1-x} - \frac{1-y}{y} F_\theta(x, y). \tag{2}$$

Bugeaud and Laurent's Theorem 1.1 applies to these nonzero algebraic x, y [5, authors' Online First PDF, p. 3]: $|x| < 1$ and $|xy^\theta| = p^{-2} < 1$; the zero-shift case goes back to Loxton and van der Poorten [5, 6]. Thus A is transcendental. The displayed affine and quadratic polynomials are nonconstant, so an algebraic value of either would force A to be algebraic. $\square$

The two-prime reduction expresses both sums through one value. The third prime leaves the binary carry of Section 2. Its rank theorem concerns that function; the scalar arithmetic depends on the multiplicities introduced next. The de-duplicated irrationality assertion is historical [3].

4 Exact multiplicities and normalised tails

Proposition 4.1 (the running least common multiple). *Let p, q, r be pairwise distinct primes and $x \geq 1$. Then the smooth-prefix LCM equals the three-prime height:* $\mathrm{L}(x) = \mathrm{H}(x)$.

Proof. Every smooth $n \leq x$ has prime exponents bounded by the corresponding integer logarithms, so $n \mid \mathrm{H}(x)$. Conversely the three maximal pure powers occur among those smooth numbers; their product divides the running LCM because they are pairwise coprime. □

The statement is smooth prefix lcm equals three-prime height. The same definition gives

$$x^3/(pqr) < \mathrm{H}(x) \leq x^3. \tag{3}$$

Both inequalities are cubic height bounds.

Proposition 4.2 (smooth-prefix LCM cells). *If $x, y \geq 1$ lie in the same logarithmic cell, then* $\mathrm{L}(x) = \mathrm{L}(y)$*: the smooth-prefix LCM cell is constant. Advancing exactly one logarithm multiplies the running value by the corresponding first-channel step, second-channel step or third-channel step. The jump set with the origin, formed from the first n positive powers in each channel together with* 1*, has cardinality* $3n + 1$.

Proof. Cell constancy and the three jump ratios follow from Proposition 4.1, since the height depends on x only through the three integer logarithms. Within one channel the first n positive powers are distinct; across channels a common value would be a positive power of two distinct primes; and 1 is not a positive prime power. □

Proposition 4.3 (height-fibre regrouping). *If $F(H)$ is a height fibre in a finite exponent box $\mathcal{B}$, the height-fibre regrouping gives*

$$\sum_{(i,j,k)\in\mathcal{B}} \mathrm{K}(i,j,k) = \sum_{H} \frac{\#F(H)}{H}. \tag{4}$$

Proof. Each term in $F(H)$ equals $1/H$. The multiplicities remain present in every subsequent infinite sum. □

The regrouping is finite smooth kernel sum grouped by height.

From now on let $P = \{2, 3, 5\}$ and $S = \mathcal{R}_P$. For $a \geq 0$ define

$$s_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{1}{\mathrm{H}(x)}, \quad T_a = \sum_{j\geq 0} s_{a+j}, \quad h_a = \frac{\mathrm{H}(2^a)}{2}, \quad X_a = h_a T_a.$$

Thus T_a is the raw tail and X_a its normalised state. The literal radix and forcing digit are

$$b_a = \frac{\mathrm{H}(2^{a+1})}{\mathrm{H}(2^a)}, \qquad m_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{\mathrm{H}(2^{a+1})}{2\,\mathrm{H}(x)}. \tag{5}$$

Lemma 4.4 (integer forcing and the four radices)**.** *For every $a \geq 0$, m_a is a positive integer and the four-letter dyadic alphabet satisfies $b_a \in \{2, 6, 10, 30\}$.*

Proof. If $x < 2^{a+1}$, the two-exponent of $\mathrm{H}(x)$ is at most a, while the other exponents are bounded by those of $\mathrm{H}(2^{a+1})$. Hence $2\,\mathrm{H}(x) \mid \mathrm{H}(2^{a+1})$. Each forcing summand is an integer, and the shell contains 2^a. Between consecutive two-powers, internal-power uniqueness leaves at most one three-power and at most one five-power. Their optional factors, together with the terminal factor two, give the four radices. □

Both clauses are literal integer forcing and four radices.

Proposition 4.5 (the actual recurrence and a polynomial bound)**.** *The series defining S, T_a converge. For every $a \geq 0$,*

$$X_{a+1} = b_a X_a - m_a, \qquad 0 < X_a \leq \frac{8640}{343}(a+1)^2 < 90(a+1)^2.$$

For every integer $B \geq 1$, either some BX_a is integral and all later states are integral, or $\mathrm{dist}(BX_a, \mathbb{Z}) \geq 1/31$ *at arbitrarily large indices.*

Proof. A dyadic shell contains at most one two-exponent for each pair of three- and five-exponents. Each of the latter lies between 0 and a, so the quadratic shell bound gives at most $(a+1)^2$ terms. By (3), $s_a \leq 30(a+1)^2/8^a$. Therefore

$$X_a \leq 15(a+1)^2 \sum_{j \geq 0} \frac{(j+1)^2}{8^j} = \frac{8640}{343}(a+1)^2.$$

This proves convergence and the bound. Splitting the first shell gives the recurrence because $m_a = h_{a+1} s_a$ and $h_{a+1} = b_a h_a$. All four clauses are short actual orbit.

Put $Y_a = BX_a$. If all sufficiently late distances are strictly below $1/31$, write $Y_a = z_a + e_a$ with $z_a \in \mathbb{Z}$ and $|e_a| < 1/31$. Then $e_{a+1} - b_a e_a$ is an integer of absolute value less than one, so $e_{a+1} = b_a e_a$. The factors $b_a \geq 2$ force a bounded such error to be zero. An integral state propagates by the integer recurrence. □

The constant above is a simple bound for the cap used below. The stronger source estimate $\widetilde{Q}(n) = (1210n^2 + 9130n + 18847)/11979$ uses the exclusion of three successive two-jumps and has a separate ordinary proof. The present cap suffices for the criterion; the sharper argument belongs in the extended record rather than in this proof.

Theorem 4.6 (actual rationality-to-reduced-carry bridge)**.** *If $S = A/D$ in lowest terms, where $D = 2^u 3^v 5^w B$ and $\gcd(B, 30) = 1$, then for every $a \geq a_0 = u + 1 + 2v + 3w$,*

$$d_a = BX_a \in \mathbb{Z}_{>0}, \qquad d_{a+1} = b_a d_a - Bm_a, \qquad d_a \leq 90B(a+1)^2.$$

Proof. For $a \geq 1$, strict boundary clearing gives

$$X_a = h_a S - Z_a, \qquad Z_a = \sum_{\substack{x \text{ smooth} \\ x < 2^a}} \frac{h_a}{\mathrm{H}(x)} \in \mathbb{Z}. \tag{6}$$

The two-exponent of h_a is $a - 1$. Moreover $a \geq 2v$ gives $2^a \geq 3^v$, and $a \geq 3w$ gives $2^a \geq 5^w$. Thus $2^u 3^v 5^w \mid h_a$ after the stated onset, and $BX_a = h_a A/(2^u 3^v 5^w) - BZ_a$ is integral. Positivity, the recurrence and the bound follow from Proposition 4.5. □

The statement is short fixed split bridge.

5 A window test and the remaining arithmetic

The actual digits in (5) define

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \qquad W_{\ell,h+1} = b_{\ell+h} W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h} F_{\ell,h} + m_{\ell+h}.$$

Induction on h gives

$$X_{\ell+h} = W_{\ell,h} X_\ell - F_{\ell,h}. \tag{7}$$

Let $\mathrm{lpr}_W(t)$ be the representative in $\{1, \dots, W\}$ of $t \bmod W$, representing a zero residue by W. Throughout this section use the valid cap

$$K(B,a) = 90B(a+1)^2. \tag{8}$$

Lemma 5.1 (finite endpoint obstruction)**.** *If d is a positive integer with $d \le K$ and $d \equiv -BF \pmod{W}$, where $W \ge 1$, then $\mathrm{lpr}_W(-BF) \le K$.*

Proof. Every positive representative of the residue is at least its least positive representative, so $\mathrm{lpr}_W(-BF) \le d \le K$. □

The statement is finite endpoint obstruction.

Theorem 5.2 (the actual window criterion)**.** *The number S is irrational if and only if*

$$\begin{gathered} \textit{for every } B \ge 1 \textit{ with } \gcd(B,30) = 1 \textit{ and every } a_0 \ge 1, \\ \textit{there are } \ell \ge a_0,\ h \ge 1 \textit{ such that } \mathrm{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > K(B, \ell+h). \end{gathered} \tag{9}$$

Proof. If S is rational, Theorem 4.6 supplies a positive integral carry $d_a = BX_a$ after its onset. Choose a window in (9) beyond that onset. Equation (7) gives $d_{\ell+h} \equiv -BF_{\ell,h} \pmod{W_{\ell,h}}$, contradicting Lemma 5.1 and the valid cap.

Conversely, let S be irrational and fix $B, \ell \ge 1$. Equation (6) makes BX_ℓ nonintegral. Put $\delta = \lceil BX_\ell \rceil - BX_\ell \in (0,1)$. Since $W_{\ell,h} \ge 2^h$ and $X_{\ell+h} = O((\ell+h+1)^2)$, for all sufficiently large h the number $\delta + BX_{\ell+h}/W_{\ell,h}$ lies in $(0,1)$. The window identity then gives the exact equality

$$\mathrm{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \delta W_{\ell,h} + BX_{\ell+h}.$$

Its exponentially growing first term eventually exceeds $K(B, \ell+h)$. This proves (9), in fact from each fixed start. □

The equivalence, at the literal smooth-number series, is short window equivalence.

The validity of the cap is needed in the rational direction; an upper polynomial growth condition by itself would not suffice. The zero cap, for example, makes positive-residue escape automatic. The criterion identifies the target in window coordinates. The unresolved point is the source-specific exclusion below.

Problem 5.3 (exact nonintegrality of every reduced tail)**.** Prove that for every $a \ge 1$ and every integer $B \ge 1$ coprime to 30,

$$BX_a \notin \mathbb{Z}. \tag{10}$$

Problem 5.4 (actual cofinal local-window escape)**.** Prove the displayed quantifier order

$$\forall B \geq 1\ (\gcd(B, 30) = 1),\ \forall a_0 \geq 1,\ \exists \ell \geq a_0\ \exists h \geq 1 : \quad \mathrm{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > K(B, \ell + h).$$

Problem 5.5 (function-faithful two-dimensional representation)**.** Express $\mathcal{D}_{2,3,5}$ as a nonconstant algebraic combination of values of a specified two-dimensional Hecke–Mahler, cone-generating or multivariate Mahler function and verify every hypothesis of a published value theorem; or give a conditional theorem under an explicit logarithmic nondegeneracy hypothesis; or prove that the literal series has no representation in the specified finite-dimensional class.

By (6), one integral value would make S rational; the bridge proves the reverse implication after clearing its denominator. Thus the pointwise formulation is equivalent to the irrationality question.

The information the remaining argument must use. The bounded-radix alternative permits the integral branch. Infinite rank and the modular minors of Corollary 2.2 constrain the kernel, while a scalar argument must also use its exact multiplicities. In the existing three-channel rigidity result, preservation of every complete same-channel block makes a perturbation a difference of three channel potentials. Two zero anchor transitions make that potential constant. A bounded rationalising lift does not automatically supply those unweighted block identities: its actual block defect is weighted. A proof using this route must establish that extra source-faithful condition. The extended record retains the exact countermodels and the other proposed representations at their stated scopes.

Statements and declarations

Proof sources. The two-prime argument uses the cited Hecke–Mahler value theorem and is Steve Fan's; it is not a Lean theorem. The arbitrary-order kernel theorem, the actual recurrence and the rationality bridge are ordinary mathematics in this paper; source locations are recorded below. Comparator records the $(2, 3, 5)$ minor $-1/15$, the running-LCM height identity, and a conditional carry-escape consumer, not the arbitrary-order rank theorem. The finite cut-rank argument and the modular corollary are proved above; no additional formal verification claim is made for the latter. The actual-series reduction is given; the source-specific cofinal escape remains unproved.

Artefact and data availability. Each source link is pinned to its own immutable revision. The exact-reference inventory records those revisions and their bytewise comparison with the current public source commit. The extended reasoning record and the accompanying source retain the detailed shell bounds, finite experiments and alternative representations with their exact hypotheses. Supplied later-source labels and public links remain separate source coordinates.

Funding and competing interests. This work received no external funding. The author declares no competing interests.

Acknowledgements. The problem numbering and status follow the Erdős Problems catalogue maintained by Thomas Bloom [4].

A Guide to the formal sources

The formal sources are recorded by the following per-link immutable revisions.

smooth3 val, three prime height, three prime kernel q, smooth prefix exponents, smooth prefix lcm, smooth3 val dvd three prime height of mem, smooth prefix lcm dvd three prime height, pure first mem smooth prefix exponents, pure second mem smooth prefix exponents, pure third mem smooth prefix exponents, smooth-prefix LCM, same three prime log cell, three prime height eq of same log cell, smooth-prefix LCM cell, three prime kernel q eq of same log cell, positive prime powers, positive prime powers card, one not mem positive prime powers, positive prime powers disjoint, three prime positive jump set, three prime positive jump set card, three prime jump set with origin, jump set with the origin, three prime height first log step, three prime height second log step, three prime height third log step, first-channel step, second-channel step, third-channel step, smooth exponent box, smooth point height, smooth height fiber, smooth height fiber kernel sum, height-fibre regrouping, three prime height le cube, kernel 235 origin, kernel 235 two, kernel 235 three, kernel 235 six, kernel 235 not rank one, tail state step, tail state step eq, smooth exponent shell, exponent unique in short interval, smooth exponent shell card le drop first, smooth exponent shell card le drop third, sorted pair quadratic, quadratic shell bound, dyadic internal power, internal-power uniqueness, dyadic block base235, four-letter dyadic alphabet, dyadic block base235 mem interval, least positive residue, least positive residue pos le, least positive residue mod eq, residue escapes window, no bounded positive state of residue escape, residue le bound of bounded positive state, no bounded positive int state of least positive residue, window base, window forcing, affine recurrence window, window forcing const mul, integral carry window, smooth3 val dvd three prime height of le, integral carry cancel common factor, reduced carry pos le of common factor, reduced integral carry window, cofinal local window escape, no positive reduced carry of cofinal local window escape, smooth lattice value, running least common multiple, pure-power height, lattice kernel, smooth prefix index set, running-lcm identity, divisibility into the height, first, second, third, cubic majorant, cell relation, cell constancy of the running value, cell constancy of the height, cell constancy of the kernel, first, second, third, first height step, second, third, positive jump count, jump count with the origin, channel cardinality, channel disjointness, exclusion of the origin, positive power sets, exponent box, height fibre, height-fibre normal form, fibre sum, point height, variable-base tail step, that names its expanded form, smooth exponent shell, short-interval uniqueness, third-coordinate projection, first-coordinate projection, quadratic shell bound, sorted quadratic estimate, non-separation witness, origin, value at two, value at three, value at six, rank-two certificate, checked internal-power uniqueness lemma, dyadic block base, exact four-case alphabet, bounded-radix consequence, checked absorption, checked cancellation, checked bound transfer, checked window transfer, window base, window forcing, checked window identity, canonical representative, positive range, congruence to the source integer, cofinal local-window escape condition, finite contradiction, integer form of the contradiction, exact least-positive-residue classifier, reduced-carry extinction theorem, absorbed-carry extinction theorem, checked, checked, residue–coboundary form, symbolic realisation, checked, carry interval, digit interval, potential classifier, one-index extinction, uniform extinction, first-block sum, four-state obstruction.

References

[1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 65. For a possibly infinite prime set Q, the page states the infinite-Q irrationality and asks what happens for finite Q with more than one element.

[2] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[3] P. Erdős, *Letter to the Editor* (written 1 January 1973), Fibonacci Quart. **12** (1974), no. 4, p. 335. The letter poses the full series as a conjecture and asserts irrationality after retaining only the distinct running-LCM values.

[4] T. F. Bloom, *Erdős Problem #269*, `erdosproblems.com/269`, accessed 28 July 2026 (page displays "last edited 28 December 2025"). The current record labels the finite-support problem open, cites `[ErGr80, p. 65]` and `[Er88c, p. 106]`, routes to the 1974 letter on p. 335, records the infinite-prime and de-duplicated variants, and explicitly describes its status as the website owner's present assessment rather than a literature-completeness guarantee.

[5] Y. Bugeaud and M. Laurent, *Transcendence and continued fraction expansion of values of Hecke–Mahler series*, Acta Arith. **209** (2023), 59–90, doi:10.4064/aa220323-18-1; authors' Online First PDF, arXiv:2203.12901v1. Theorem 1.1 is on p. 3 of the inspected Online First PDF; the journal publication is pp. 59–90. The authors note there that its $\rho = 0$ case was already obtained by Loxton and van der Poorten.

[6] J. H. Loxton and A. J. van der Poorten, *Arithmetic properties of certain functions in several variables III*, Bull. Austral. Math. Soc. **16** (1977), 15–47.

[7] S. Fan, comment on Erdős Problem #269, erdosproblems.com forum, thread 269, 26 June 2026. The comment gives the two-channel factorisation, the Hecke–Mahler reduction, and the transcendence conclusion for $|P| = 2$; follow-up comments there note the extension to arbitrary coprime pairs.

Companion system context. The claim and trust boundary, cold-clone route to proof authority, and public contribution protocol are described in sibling papers. Those descriptions do not change the mathematical status of this note.